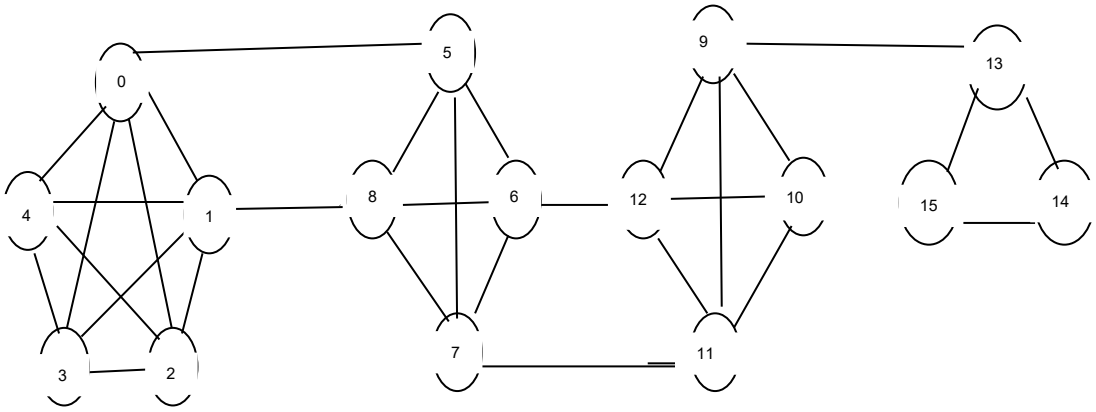
Project 2

Write a C++ program to apply either Spectral Clustering or Markov techniques for finding the clusters of the graph depicted below. Python or C++ languages are allowed. It is recommended to implement the algorithms using the linear algebra packages from the standard libraries of C++ or NumPy library in Python. Just recall that for the Spectral Clustering you need to calculate the eigenvalues (the first 2) and apply k-means of the first two eigenvectors. For Spectral Clustering find 2-means (two clusters), 3-means (three clusters), and 4-means (four clusters). In the case you chose to implement Markov Clustering remember that the technique automatically finds the clusters. Must show some of the intermediate calculations. The output should specify the number of clusters chosen and the vertices set corresponding to each cluster.

Input graph